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GATE ECE · ANALOG CIRCUITS · CHAPTER-WISE PYQS

Frequency Response

GATE ECE Analog Circuits · Previous Year Questions with Answer Key

19	14	1993 – 2025	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	11 questions
Marks 2	7 questions
Marks 5	1 question
Answer key	at the end

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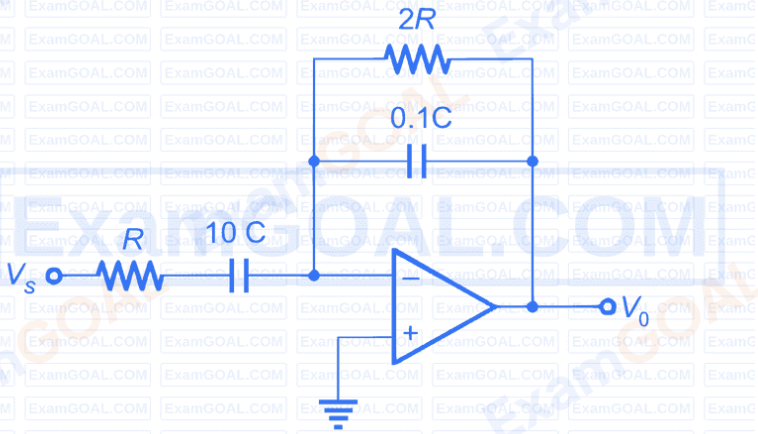
1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Marks 1 11 questions

Q1. GATE ECE 2025

All the components in the bandpass filter given below are ideal. The lower -3 dB frequency of the filter is 1 MHz . The upper -3 dB frequency (in MHz, rounded off to the nearest integer) is _____ .

Open solution



Q2. GATE ECE 2017 Set 1

The Miller effect in the context of a Common Emitter amplifier explains

(A) an increase in the low-frequency cutoff frequency

(B) an increase in the high-frequency cutoff frequency

(C) a decrease in the low-frequency cutoff frequency

(D) a decrease in the high-frequency cutoff frequency

Open solution

Q3. GATE ECE 2016 Set 2

Which one of the following statements is correct about an ac-coupled common- emitter amplifier operating in the mid-band region?

(A) The device parasitic capacitances behave like open circuits, whereas coupling and bypass capacitances behave like short circuits.

(B) The device parasitic capacitances, coupling capacitances and bypass capacitances behave like open circuits.

(C) The device parasitic capacitances, coupling capacitances and bypass capacitances behave like short circuits.

(D) The device parasitic capacitances behave like short circuits, whereas coupling and bypass capacitances behave like open circuits.

Open solution

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Q4. GATE ECE 2003

Generally, the gain of a transistor Amplifier falls at high frequency due to the

(A) Internal Capacitances of the device

(B) Coupling Capacitor at the input

(C) Skin Effects

(D) Coupling Capacitor at the output

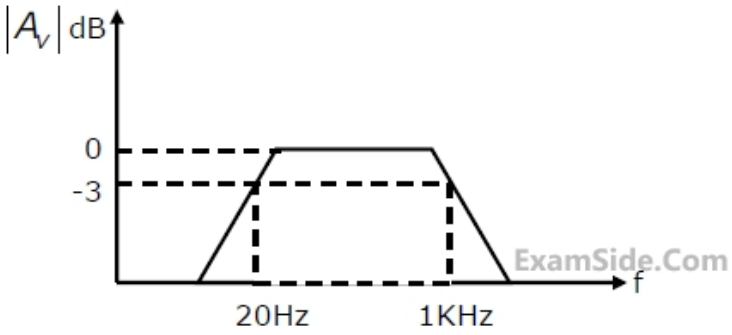
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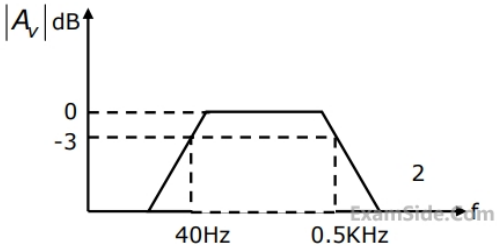
Q5. GATE ECE 2002

Three identical RC-Coupled transistor amplifiers are cascaded. If each of the amplifiers has a frequency response as shown in figure, the overall frequency response is as given as

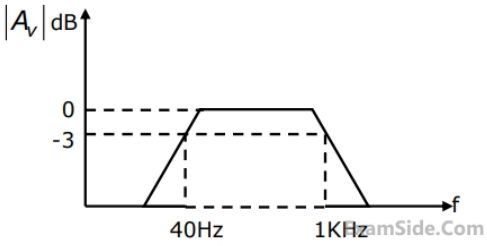
Open solution



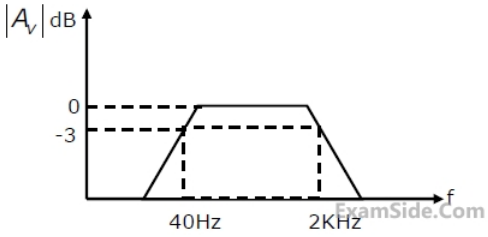
(A)



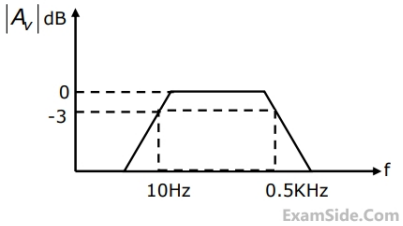
(B)



(C)



(D)



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Q6. GATE ECE 2000

The current gain of a bipolar transistor drops at high frequencies because of

(A) Transistor capacitances

(B) High current effects in the base

(C) Parasitic inductive elements

(D) The Early effect

Open solution

Q7. GATE ECE 1998

The f_T of a BJT is related to its g_m, C_π and C_μ as follows

(A) $f_T = \frac{C_\pi + C_\mu}{g_m}$

(B) $f_T = \frac{2\pi(C_\pi + C_\mu)}{g_m}$

(C) $f_T = \frac{g_m}{C_\pi + C\mu}$

(D) $f_T = \frac{g_m}{2\pi(C_\pi + C_\mu)}$

Open solution

Q8. GATE ECE 1996

An npn transistor has a beta cut-off frequency f_β of **1MHz** and Common Emitter short circuit low frequency current gain β_o of 200. The unity gain frequency f_T and the alpha cut-off frequency f_α respectively are

(A) 200 MHZ, 201 MHz

(B) 200 MHZ, 199 MHz

(C) 199 MHZ, 200 MHz

(D) 201 MHZ, 200 MHz

Open solution

Q9. GATE ECE 1995

An RC-Coupled Amplifier is assumed to have a single-pole low frequency transfer function. The maximum lower-cutoff frequency allowed for the Amplifier to pass 50 Hz square wave with no more than 10% tilt is _____.

Open solution

Q10. GATE ECE 1995

An Amplifier has an open-loop gain of 100 and its lower and upper-cut-off frequency of 100 Hz and 100 kHz respectively, a feedback network with a feedback factor of 0.99 is connected to the amplifier. the lower-and upper-cut-off frequency's are at .....and .....

Open solution

Q11. GATE ECE 1993

In a multi-stage RC-Coupled Amplifier the coupling capacitor.

(A) limits the low frequency response.

(B) limits the high frequency response.

(C) does not affect the frequency response.

(D) blocks the d.c. components without effecting the frequency response.

Open solution

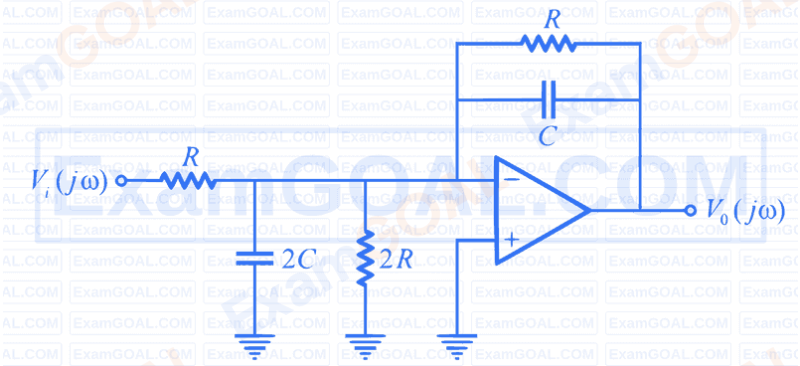
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Marks 2 7 questions

Q12. GATE ECE 2020

Open solution

The components in the circuit given below are ideal. If $R = 2\text{k}\Omega$ and $C = 1\mu\,\text{F}$, the -3 dB cut-off frequency of the circuit in Hz is



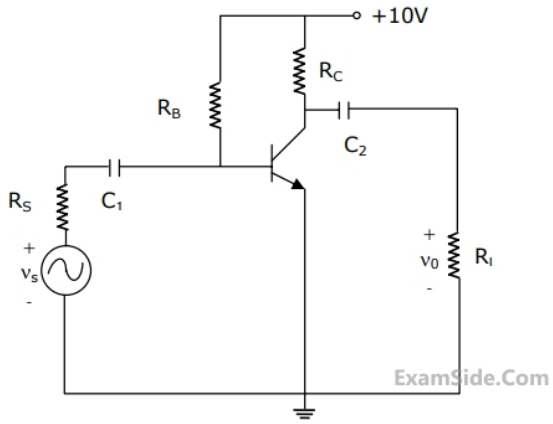
(A) 79.58 (B) 14.92 (C) 59.68 (D) 34.46

Q13. GATE ECE 2010

Open solution

Consider the common emitter amplifier shown below with the following circuit parameters:

$\beta = 100,\ g_m = 0.3861\ \text{A}/V,\ r_0 = \infty,\ r_\pi = 259\,\Omega,$
$R_s = 1\,K\Omega,\ R_B = 93\,K\Omega,\ R_C = 250\,\Omega,$
$R_L = 1\,K\Omega,\ C_1 = \infty\ and\ C_2 = 4.7\,\mu F.$



The Resistance seen by the source Vs is

(A) 258Ω (B) 1258Ω (C) 93Ω (D) 100Ω

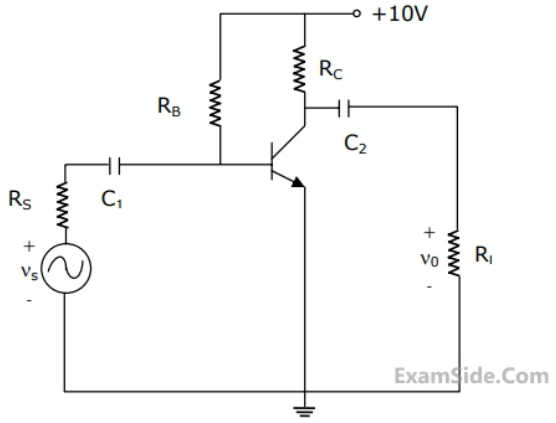
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Q14. GATE ECE 2010

Open solution

Consider the common emitter amplifier shown below with the following circuit parameters:

$\beta = 100,\ g_m = 0.3861\ \mathrm{A}/V,\ r_0 = \infty,\ r_\pi = 259\,\Omega,$
$R_s = 1\,K\Omega, R_B = 93\,K\Omega,\ R_C = 250\,\Omega,$
$R_L = 1\,K\Omega,\ C_1 = \infty\ and\ C_2 = 4.7\,\mu F.$



The lower cut-off frequency due to C_2 is

(A) 33.9 Hz (B) 27.1 Hz (C) 13.6 Hz (D) 16.9 Hz

Q15. GATE ECE 2001

Open solution

An npn BJT has g_m = 38 mA/V, $C_\mu = 10^{-14}$ F, $C_\pi = 4 \times 10^{-13}\,F$ and DC current gain $\beta = 90$. For this transistor f_T and f_β are

(A) $f_T = 1.64 \times 10^8\ Hz\,and$
$f_\beta = 1.47 \times 10^{10}\ Hz$

(B) $f_T = 1.47 \times 10^{10}\ Hz\,and$
$f_\beta = 1.64 \times 10^8\ Hz$

(C) $f_T = 1.33 \times 10^{12}\ Hz\,and$
$f_\beta = 1.47 \times 10^{10}\ Hz$

(D) $f_T = 1.47 \times 10^{10}\ Hz\,and$
$f_\beta = 1.33 \times 10^{12}\ Hz$

Q16. GATE ECE 1999

Open solution

An NPN transistor (with $C_\pi = 0.3$ pf) has a unity gain cutoff frequency f_T of 400 MHz at a DC-bias current I_c = 1mA. The value of its C_μ (in pf) is approximately $(V_T = 26\,mV)$.

(A) 15 (B) 30 (C) 50 (D) 96

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Q17. GATE ECE 1999

An amplifier is assumed to have a single pole high frequency transfer function. The rise time of its output response to a step function input is 35 nsec. The upper 3 dB frequency (in MHz) for the amplifier to a sinusoidal input is approximately at

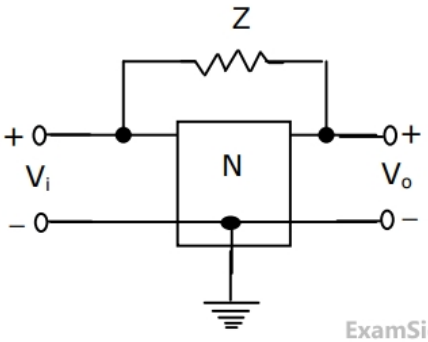
Open solution

(A) 4.55 (B) 10 (C) 20 (D) 28.6

Q18. GATE ECE 1996

In the circuit shown in Fig. is a finite gain amplifier with a gain of K, a very large input impedance, and a very low output impedance. The input impedance of the feedback amplifier with the feedback impedance Z conducted as shown will be

Open solution



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(A) $Z\left[1-\frac{1}{K}\right]$ (B) $Z(1-K)$

(C) $\left[\frac{Z}{K-1}\right]$ (D) $\left[\frac{Z}{1-K}\right]$

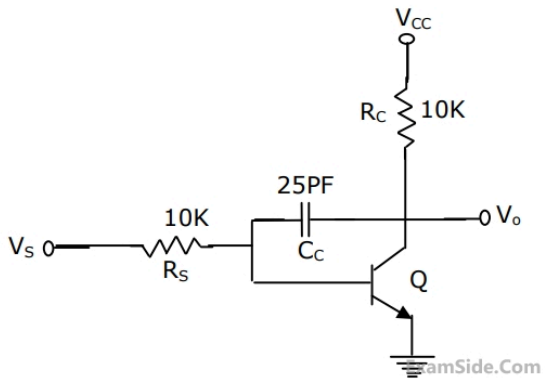
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Marks 5 1 questions

Q19. GATE ECE 1996

Open solution

A common-emitter amplifier with an external capacitor C_C connected across the base and the collector of the transistor is shown in Fig.



Transistor data g_m = 5mA/V, $r_\pi = 200K\Omega$, $C_\pi = 1.5$ pF and $C\mu = 0.5$ pF determine the upper cutoff frequency f_H of the amplifier

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Answer Key

Marks 1

1. **50**	2. **D**	3. **A**	4. **A**	5. **B**
6. **A**	7. **D**	8. **D**	9. **1.59 Hz**	10. **10 MHz, 1 Hz**
11. **A**				

Marks 2

12. **A**	13. **A**	14. **B**	15. **B**	16. **A**
17. **B**	18. **D**			

Marks 5

19. **f_H= 18.345 KHz**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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